

DREAM6
A Unified Spectral–Dynamical Theory of SAT Witness Extraction
with Residual Dynamics, IPC Geometry,
Annihilation Operators, and Repair Kernels

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1 Central Statement

DREAM6 is a deterministic spectral–dynamical framework for SAT witness extraction.

The construction combines:

- (i) smooth residual SAT dynamics,
- (ii) deterministic holographic phase embeddings,
- (iii) bounded-overlap Gram operators,
- (iv) invariant phase certifiers (IPC),
- (v) annihilation operators,
- (vi) residual transducers,
- (vii) localized repair kernels.

The implemented computational spine is

$$F \mapsto (Z, G_\Phi, s, w) \mapsto IPC(Z) \mapsto \hat{x} \mapsto \text{repair} \mapsto x^\star.$$

The target theorem is:

$$(A1 - A5) \implies 3SAT \in P \implies P = NP.$$

The purpose of this document is to formalize the entire mathematical structure used by the implemented DREAM6 operator solver.

2 Input Object and Verification Layer

Let

$$F = (C_1, \dots, C_M)$$

be a CNF formula over Boolean variables

$$x_1, \dots, x_N \in \{0, 1\}.$$

Each clause is

$$C_j = (\ell_{j1}, \dots, \ell_{jk_j}), \quad \ell_{jr} \in \{\pm 1, \dots, \pm N\}.$$

Literal satisfaction is defined by

$$\text{sat}(\ell, x) = \begin{cases} x_{|\ell|}, & \ell > 0, \\ 1 - x_{|\ell|}, & \ell < 0. \end{cases}$$

The clause predicate is

$$C_j(x) = 1 \iff \exists r : \text{sat}(\ell_{jr}, x) = 1.$$

The verifier is

$$\text{Unsat}(F, x) = \sum_{j=1}^M (1 - C_j(x)).$$

The SAT witness objective is:

$$\text{Find } x \in \{0, 1\}^N \text{ such that } \text{Unsat}(F, x) = 0.$$

Verification complexity:

$$\text{Verify}(x) = O(M).$$

SAT certificates are therefore immediate.

UNSAT is not inferred from failure of the dynamics.

3 Boolean Spin Geometry

Boolean variables are encoded through spins

$$s_i \in \{-1, +1\}, \quad x_i = \frac{1 + s_i}{2}.$$

For a literal pair

$$(i, b), \quad b \in \{+1, -1\},$$

we define:

$$b = +1 \iff x_i, \quad b = -1 \iff \neg x_i.$$

The smooth literal value is

$$v_{i,b}(s) = \frac{1 + bs_i}{2}.$$

Thus

$$v_{i,+1}(s) = \frac{1 + s_i}{2},$$

$$v_{i,-1}(s) = \frac{1 - s_i}{2}.$$

On Boolean vertices:

$$v_{i,b}(s) = 1 \iff (i, b) \text{ is true.}$$

4 Clause Residual Geometry

For a clause

$$C = ((i_1, b_1), (i_2, b_2), (i_3, b_3))$$

define the residual

$$R_C(s) = \prod_{k=1}^3 (1 - v_{i_k, b_k}(s)).$$

Equivalently:

$$R_C(s) = \frac{1}{8} \prod_{k=1}^3 (1 - b_k s_{i_k}).$$

On Boolean vertices:

$$R_C(s) = 1 \iff C \text{ is false,}$$

$$R_C(s) = 0 \iff C \text{ is true.}$$

The global residual energy is

$$E_F(s) = \sum_{C \in F} R_C(s).$$

On Boolean vertices:

$$E_F(s) = \text{Unsat}(F, x(s)).$$

5 Gradient Structure

For variable s_i ,

$$\partial_{s_i} R_C(s) = -\frac{b_i}{8} \prod_{k \neq i} (1 - b_k s_k).$$

Hence

$$\nabla E_F(s) = \sum_{C \in F} \nabla R_C(s).$$

Critical points satisfy

$$\nabla E_F(s) = 0.$$

6 Boolean Pinning Potential

Define the double-well pinning:

$$V_\mu(s) = \mu \sum_{i=1}^N (s_i^2 - 1)^2, \quad \mu > 0.$$

Properties:

$$V_\mu(s) = 0 \iff s \in \{-1, +1\}^N.$$

Gradient:

$$\partial_{s_i} V_\mu(s) = 4\mu s_i (s_i^2 - 1).$$

The total DREAM6 potential is

$$U_F(s) = E_F(s) + V_\mu(s).$$

7 BOOM-Style Phase Coordinate

Introduce a global phase

$$\theta(t) \in \mathbb{R}/2\pi\mathbb{Z}.$$

The DREAM6 Lagrangian is

$$\mathcal{L} = \frac{1}{2}\|\dot{s}\|^2 - (E_F(s) + V_\mu(s)) + \frac{\kappa}{2}\dot{\theta}^2 + \eta\dot{\theta}E_F(s) - \zeta \cos \theta.$$

The coupling

$$\eta\dot{\theta}E_F(s)$$

is the BOOM-style presence interaction.

8 Euler–Lagrange Dynamics

Spin dynamics:

$$\ddot{s}_i + \partial_{s_i}E_F(s) + \partial_{s_i}V_\mu(s) - \eta\dot{\theta}\partial_{s_i}E_F(s) = 0.$$

Equivalent vector form:

$$\ddot{s} = -\nabla V_\mu(s) - (1 - \eta\dot{\theta})\nabla E_F(s).$$

Phase dynamics:

$$\kappa\ddot{\theta} + \zeta \sin \theta + \eta \frac{d}{dt}E_F(s) = 0.$$

9 Damped Residual Dynamics

The implemented solver introduces damping:

$$\ddot{s} + \gamma_s \dot{s} + \nabla V_\mu(s) + (1 - \eta\dot{\theta})\nabla E_F(s) = 0,$$

$$\kappa\ddot{\theta} + \gamma_\theta \dot{\theta} + \zeta \sin \theta + \eta \frac{d}{dt}E_F(s) = 0.$$

10 Lyapunov Functional

Define

$$\mathcal{V}(t) = E_F(s) + V_\mu(s) + \frac{1}{2}\|\dot{s}\|^2 + \frac{\kappa}{2}\dot{\theta}^2 + \zeta(1 - \cos \theta).$$

Under damping:

$$\frac{d}{dt}\mathcal{V}(t) \leq 0.$$

Thus DREAM6 dynamics descends toward:

$$E_F(s) = 0$$

or toward a low-residual localized attractor.

11 Zero-Event Geometry

Define the residual critical set

$$\mathcal{Z} = \{t : \nabla E_F(s(t)) = 0\}.$$

Define the exact collapse set

$$\mathcal{Z}_0 = \{t : E_F(s(t)) = 0\}.$$

Define clause events

$$\mathcal{Z}_C = \{(t, j) : R_{C_j}(s(t)) = 0\}.$$

Switching rules:

$$E_F(s(t)) = 0 \implies \text{VERIFY SAT},$$

$$\nabla E_F(s(t)) \approx 0 \quad \text{and} \quad E_F(s(t)) > 0 \implies \text{POLISH/REPAIR}.$$

12 Prime Lock Windows

Fix parameters

$$T = \text{time horizon}, \quad m = \text{lock length}.$$

Choose:

$$p \geq M$$

prime,

$$a \not\equiv 0 \pmod{p}, \quad b \in \{0, \dots, p-1\}.$$

Define clause offsets:

$$o_j = \left\lfloor T \cdot \frac{(aj + b) \bmod p}{p} \right\rfloor.$$

Lock window:

$$L_j = \{o_j, \dots, o_j + m - 1\} \pmod{T}.$$

Lock fraction:

$$\rho_{lock} = \frac{m}{T}.$$

13 Walsh–Hadamard Carrier Layer

Define

$$m_{eff} = 2^{\lceil \log_2 m \rceil}.$$

Each clause receives a Hadamard mask

$$s_j \in \{\pm 1\}^{m_{eff}}$$

restricted to length m .

Normalization:

$$\|s_j\|_2 = \sqrt{m}.$$

Mis-phase injection flips

$$k = \lfloor \zeta_0 m \rfloor$$

positions.

The mask correlation slack satisfies

$$\varepsilon(m) \leq \frac{2}{\sqrt{m_{eff}}} + \frac{2}{m}.$$

14 Phase Matrix

Define

$$Z \in \mathbb{C}^{T \times M}.$$

Entries:

$$Z_{tj} = \begin{cases} g_j s_{j,t-o_j}, & t \in L_j, \\ -1, & t \notin L_j. \end{cases}$$

Projection onto the unit circle:

$$\Pi_{S^1}(z) = \frac{z}{|z|}.$$

Applied componentwise:

$$Z \leftarrow \Pi_{S^1}(Z).$$

Hence

$$|Z_{tj}| = 1.$$

15 Clause Graph

Define the clause graph

$$G_\Phi = (V, E), \quad V = [M].$$

Edges:

$$(i, j) \in E \iff C_i, C_j \text{ share a variable.}$$

Bounded degree condition:

$$\deg(i) \leq d = O(1).$$

16 Signed Coupling Dynamics

Define clause gauges

$$g_j \in \{\pm 1\}.$$

Signed coupling:

$$\sigma_{ij} = g_i g_j.$$

Update rule:

$$Z_i \leftarrow \Pi_{S^1} \left((1 - \eta_{ij}) Z_i + \eta_{ij} \sigma_{ij} Z_j + h \right).$$

Residual phase:

$$R_{ij} = \text{wrap}_\pi \left(\arg(Z_i \sigma_{ij} Z_j) \right).$$

Soft inverse residual weight:

$$w_{\text{soft}} = \frac{1}{\sqrt{R_{ij}^2 + \varepsilon_{\text{dyn}}^2}}.$$

17 Clause Gram Operator

Define the time-averaged Gram operator:

$$G_{\text{clause}} = \frac{1}{T} Z^* Z.$$

Componentwise:

$$G_{\text{clause}}(i, j) = \frac{1}{T} \sum_{t=1}^T \overline{Z_{ti}} Z_{tj}.$$

Diagonal:

$$G_{\text{clause}}(i, i) = 1.$$

18 Cross-Term Attenuation

Define

$$\kappa = (1 - 2\zeta_0)^2 + \varepsilon(m) + \frac{1}{T}.$$

Then

$$\max_i \sum_{j \in \mathcal{N}(i)} |G_{clause}(i, j)| \leq d\kappa.$$

Time decomposition:

$$[1, T] = \Omega_{lock} \cup \Omega_{mixed} \cup \Omega_{out}.$$

Then

$$G_{clause}(i, j) = A_{lock} + A_{mixed} + A_{out}.$$

Lock contribution:

$$|A_{lock}| \leq (1 - 2\zeta_0)^2 + \frac{1}{\sqrt{m_{eff}}}.$$

Mixed contribution:

$$|A_{mixed}| \leq \frac{2}{m}.$$

Outside contribution:

$$|A_{out}| \leq \frac{1}{T}.$$

Hence:

$$|G_{clause}(i, j)| \leq (1 - 2\zeta_0)^2 + \frac{2}{\sqrt{m_{eff}}} + \frac{2}{m} + \frac{1}{T}.$$

19 Gershgorin Spectral Control

By Gershgorin:

$$\lambda(G_{clause}) \subset \bigcup_i B \left(G_{ii}, \sum_{j \neq i} |G_{ij}| \right).$$

Thus off-diagonal perturbation remains bounded by

$$d\kappa.$$

20 Annihilation Operator

Define

$$D = \text{diag}(G_{\text{clause}}), \quad N = G_{\text{clause}} - D.$$

The annihilation operator is

$$\mathcal{A}_\lambda(G) = G - \lambda \Pi_{\text{noise}}(N).$$

Spectral noise projection:

$$\Pi_{\text{noise}}(N) = \sum_{\mu_k < \tau} \mu_k u_k u_k^*.$$

The effective operator becomes

$$G_{\text{eff}} = \mathcal{A}_\lambda(G).$$

Row-sum radar:

$$S2_i = \sum_{j \in \mathcal{N}(i)} |G_{\text{eff}}(i, j)|.$$

21 Invariant Phase Certifier

IPC produces clause phasors

$$a_j = A_j e^{i\phi_j}.$$

Global phase:

$$\theta = \arg \sum_j a_j$$

or the Z_2 -stable version

$$\theta = \frac{1}{2} \arg \sum_j a_j^2.$$

U(1) coherence:

$$\text{coh}_{U(1)} = \left| \frac{1}{M} \sum_j e^{i(\phi_j - \theta)} \right|.$$

Z_2 coherence:

$$\text{coh}_{Z_2} = \left| \frac{1}{M} \sum_j e^{2i(\phi_j - \theta)} \right|.$$

22 Z2 Ambiguity Control

Because

$$a_j \sim -a_j,$$

the naive phase readout is polarity ambiguous.

The DREAM6 correction uses

$$\theta = \frac{1}{2} \arg \sum_j a_j^2.$$

Since

$$(-a_j)^2 = a_j^2,$$

the sign ambiguity disappears.

This is the implemented DREAM6 Z_2 symmetry-breaking mechanism.

23 Variable–Clause Incidence

Define

$$I(v) = \{(j, \sigma_{jv}) : x_v \text{ occurs in } C_j\}.$$

Signs:

$$\sigma_{jv} = \begin{cases} +1, & x_v, \\ -1, & \neg x_v. \end{cases}$$

24 Z2-Aware Boolean Readout

Carrier alignment:

$$\alpha_j = \cos(\phi_j - \theta).$$

If

$$\alpha_j < 0,$$

apply the phase flip

$$\phi_j \leftarrow \phi_j + \pi.$$

Define

$$g_j = \alpha_j^2.$$

Target phase:

$$\theta_{jv} = \begin{cases} \theta + \rho_j, & \sigma_{jv} = +1, \\ \theta + \pi - \rho_j, & \sigma_{jv} = -1, \end{cases}$$

where

$$\rho_j = 0.1(1 - w_j).$$

The DREAM6 Boolean score is

$$S_v = \sum_{(j, \sigma_{jv}) \in I(v)} w_j A_j g_j \cos \left(\text{wrap}_\pi(\phi_j^* - \theta_{jv}) \right).$$

Readout:

$$\hat{x}_v = [S_v \geq 0].$$

Spin readout:

$$\hat{s}_v = \text{sign}(S_v).$$

25 Residual Transducer

The residual transducer converts spectral information into dynamical momentum.

Initial state:

$$s_v(0) = \tanh(\beta S_v).$$

Initial momentum:

$$\dot{s}_v(0) = \eta_v \tanh(\alpha S_v).$$

Local residual density:

$$R_v = \sum_{C_j \ni v} R_{C_j}(s).$$

Transducer coefficient:

$$\eta_v = \min \left(\eta_{max}, \frac{|S_v|^\gamma}{\varepsilon + R_v} \right).$$

This is the mathematical form of the implemented **residual-xducer-cap** mechanism.

The effective IPC force becomes

$$F_v^{IPC} = \eta_v \nabla_{s_v} E_F(s).$$

The resulting spin dynamics is

$$\ddot{s}_v + \gamma_s \dot{s}_v + \nabla_{s_v} V_\mu(s) + (1 - F_v^{IPC}) \nabla_{s_v} E_F(s) = 0.$$

26 Readout Stability

Suppose

$$(2x_v^* - 1)S_v \geq \gamma.$$

If perturbations satisfy

$$|\Delta_v| < \gamma,$$

then

$$\text{sign}(S_v + \Delta_v) = 2x_v^* - 1.$$

Proof:

$$(2x_v^* - 1)(S_v + \Delta_v) \geq \gamma - |\Delta_v| > 0.$$

27 IPC Margin

Define

$$\mu_{IPC} = \beta^2 \cos^2 \delta.$$

Uniform margin condition:

$$\mu_{IPC} \geq (N + M)^{-q}.$$

28 Dynamical Lock Stability

Define

$$\Lambda = \frac{Kh}{\mu(1 + d\kappa)}.$$

Stable operation requires

$$\Lambda_{min} < \Lambda < \Lambda_{max}.$$

Equivalently:

$$\mu(1 + d\kappa) < Kh < C_{chaos}\mu.$$

Interpretation:

- too small Kh : no coherent drive,
- too large Kh : unstable oscillatory chaos,
- $d\kappa$: effective spectral leakage.

29 Soft-ABS Regularization

Define

$$F_\varepsilon(x) = \sqrt{x^2 + \varepsilon^2}.$$

Derivative:

$$F'_\varepsilon(x) = \frac{x}{\sqrt{x^2 + \varepsilon^2}}.$$

Bound:

$$|F'_\varepsilon(x)| \leq 1.$$

Purpose:

- bounded singular regularization,
- stable inverse residual weights,
- no destructive division by zero.

30 Closure Integral

Define the closure phase:

$$\Theta = \frac{1}{Mm} \sum_{j=1}^M \sum_{t \in L_j} \arg(Z_{tj}).$$

Refined time:

$$\tau = t + \lambda\Theta.$$

31 Residual Polish

Residual clause set:

$$\mathcal{R}(x) = \{j : C_j(x) = 0\}.$$

Flip gain:

$$\Delta_v = \text{Unsat}(F, x \oplus e_v) - \text{Unsat}(F, x).$$

Accept flip iff

$$\Delta_v < 0.$$

Then

$$\text{Unsat}(F, x_{k+1}) < \text{Unsat}(F, x_k).$$

32 Repair Kernel

Let

$$F_r$$

be the radius- r neighborhood around unsatisfied clauses.
Variables outside F_r are locked:

$$x_v = x_v^{seed}.$$

Mapped variables:

$$y_v = 1 \iff x_v = x_v^{seed}.$$

Unmapping:

$$x_v = \begin{cases} x_v^{seed}, & y_v = 1, \\ 1 - x_v^{seed}, & y_v = 0. \end{cases}$$

Repair CNF:

$$F_{repair} = F_{mapped} \wedge \bigwedge_{v \notin F_r} (y_v).$$

Correctness:

$$F_{repair}(y) = 1 \implies F(x(y)) = 1.$$

33 Residual Localization

Uncertain set:

$$U_\tau = \{v : |S_v| \leq \tau\}.$$

Target bound:

$$|U_\tau| \leq (N + M)^r.$$

Ideal target:

$$|U_\tau| \leq O(\log^k(N + M)).$$

34 Master Computational Spine

The implemented DREAM6 chain is

$$F \mapsto Z \mapsto G \mapsto \mathcal{A}_\lambda(G) \mapsto (a_j, \theta) \mapsto S_v \mapsto (s(0), \dot{s}(0)) \mapsto \text{residual flow} \mapsto \mathcal{Z} \mapsto \text{repair kernel} \mapsto x^*$$

Core principle:

$$\text{spectral coherence} \Rightarrow \text{directed dynamical momentum} \Rightarrow \text{localized residual core}$$

35 Assumptions A1–A5

A1 — Polynomial Construction

All DREAM6 structures are computable in polynomial time.

A2 — Cross-Term Control

The spectral leakage remains bounded:

$$d\kappa \leq \text{poly}(N + M)^{-1}.$$

A3 — Uniform IPC Margin

There exists q such that

$$\mu_{IPC} \geq (N + M)^{-q}.$$

A4 — Residual Localization

Residual uncertainty satisfies

$$|U_\tau| \leq (N + M)^r.$$

A5 — Polynomial Repair

Repair succeeds in polynomial time whenever A3–A4 hold.

36 Conditional Witness Theorem

Theorem.

If A1–A5 hold uniformly for all satisfiable 3-CNF formulas, then DREAM6 is a deterministic polynomial-time SAT witness extractor.

Proof Skeleton.

1. A1 gives polynomial construction.
2. A2 preserves the coherent SAT carrier.
3. A3 guarantees inverse-polynomial IPC margin.
4. Readout stability fixes all variables outside U_τ .
5. A4 localizes the residual core.
6. A5 repairs the residual core in polynomial time.
7. The verifier confirms

$$\text{Unsat}(F, x^\star) = 0.$$

Hence satisfiable 3-CNF admits deterministic polynomial witness extraction.

37 P versus NP Consequence

If A1–A5 hold uniformly:

$$3SAT \in P.$$

Since $3SAT$ is NP-complete:

$$P = NP.$$

Current logical status:

$$(A1 - A5) \implies P = NP.$$

The framework is therefore a conditional route toward $P = NP$, pending full proof of the remaining obligations.

38 Current Open Obligations

1. Uniform IPC carrier theorem.
2. No exponentially small margin.
3. No false stable attractors.
4. Full Z_2 stability proof.
5. Polynomial residual localization.
6. Polynomial polish complexity.
7. Polynomial repair theorem.

8. Polynomial precision bounds.
9. Polynomial convergence of residual flow.
10. Formal UNSAT certificate layer.

39 Final Structural Identity

DREAM6 is the composition

CNF residual geometry + unitary phase carrier + Gram attenuation + IPC extraction + annihilation operator

with the central mechanism

spectral information retention + localized residual repair + polynomial convergence

and the exact formal status

$(A1 - A5) \implies \text{polytime SAT witness extraction} \implies P = NP.$